

This form documents the artifacts associated with the article (i.e., the data and code supporting the computational findings) and describes how to reproduce the findings.

Part 1: Data

- ☐ This paper does not involve analysis of external data (i.e., no data are used or the only data are generated by the authors via simulation in their code).
- ☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Abstract

Datasets used in the real data analysis in the paper ‘Spherically Embedded Time Series with Unknown Trend and Periodic Components’, covering U.S. energy generational compositional time series and distributional time series from the New York City City bike sharing system. The two datasets are available in the Github repository <https://github.com/d5rhtkp4kc-alt/STPD/tree/main/real%20data/dataset>. The sources of each dataset are explicitly mentioned in the paper and are as follows: <https://www.eia.gov/electricity/data/browser/> (U.S. energy generational data) and <https://citibikenyc.com/system-data> (New York City City bike sharing system data).

Availability

- ☒ Data **are** publicly available.
- ☐ Data **cannot be made** publicly available.

If the data are publicly available, see the *Publicly available data* section. Otherwise, see the *Non-publicly available data* section, below.

Publicly available data

- ☐ Data are available online at:
- ☒ Data are available as part of the paper’s supplementary material.
- ☐ Data are publicly available by request, following the process described here:
- ☐ Data are available through some other mechanism, described here:

Non-publicly available data

Description

File format(s)

- ☒ CSV or other plain text.
- ☐ Software-specific binary format (.Rda, Python pickle, etc.):
- ☐ Standardized binary format (e.g., netCDF, HDF5, etc.):
- ☐ Other (please specify):

Data dictionary

- ☐ Provided by authors in the following file(s):
- ☐ Data file(s) is (are) self-describing (e.g., netCDF, HDF5, Zarr files)
- ☒ Available at the following URL: <https://github.com/d5rhtkp4kc-alt/STPD/tree/main/real%20data/dataset>

Additional Information (optional)

The folder ‘real data/dataset’ contains two datasets: US_energy.csv and trip.csv.

1. US_energy.csv — US energy generation composition dataset downloaded from <https://www.eia.gov/energy/data/browser/>. For details, see Section 6.1 of the paper.
2. trip.csv — Preprocessed New York City Citi bike sharing system data with original dataset downloaded from <https://citibikenyc.com/system-data>. For details, see Section 6.2 of the paper.

Part 2: Code

Abstract

Code to reproduce all the numerical results in the paper ‘Spherically Embedded Time Series with Unknown Trend and Periodic Components’, available in the GitHub repository: <https://github.com/d5rhtkp4kc-alt/STPD>. It also contains Python code for TimePFN downloaded from <https://github.com/egetaga/TimePFN>.

Description

Code format(s)

- ☒ Script files
 - ☒ R
 - ☒ Python
 - ☐ Julia
 - ☐ MATLAB
 - ☐ Other:
- ☐ Package
 - ☐ R
 - ☐ Python
 - ☐ Julia
 - ☐ MATLAB toolbox
 - ☐ Other:
- ☒ Reproducible report
 - ☐ Quarto or R Markdown
 - ☒ Jupyter notebook
 - ☐ Other:
- ☐ Shell script
- ☐ Other (please specify):

Supporting software requirements

Primary software used R version 4.5.1, Python 3.13.13

Libraries and dependencies used by the code

1. **R Packages:** ggplot2 (version 3.5.2), plot3D (version 1.4.1), ggforce (version 0.5.0), patchwork (version 1.3.0), psych (version 2.5.3), osqp (version 0.6.3.3), transport (version 0.15-4), easyCODA (version 0.40.2), tidyr (version 1.3.1), movMF (version 0.2-9), trust (version 0.1-8), parallel (version 4.5.0), foreach (version 1.5.2), doSNOW (version 1.0.20), xtable (version 1.8-4).
2. **Python Packages:** Standard library modules for Python 3.13.13 — copy, os, random, sys, time. Third-party packages — numpy (version 2.4.4), pandas (version 3.0.2), torch (version 2.12.0), matplotlib (version 3.10.8), scipy (version 1.17.1).

Supporting system/hardware requirements (optional)

Parallelization used

- ☐ No parallel code used
- ☐ Multi-core parallelization on a single machine/node
 - Number of cores used:
- ☒ Multi-machine/multi-node parallelization
 - Number of nodes and cores used: 200

License

- ☒ MIT License (default)
- ☐ BSD
- ☐ GPL v3.0
- ☐ Creative Commons
- ☐ Other: (please specify)

Additional information (optional)

Part 3: Reproducibility workflow

Scope

The provided workflow reproduces:

- ☒ Any numbers provided in text in the paper
- ☒ The computational method(s) presented in the paper (i.e., code is provided that implements the method(s))
- ☒ All tables and figures in the paper
- ☐ Selected tables and figures in the paper, as explained and justified below:

Workflow

Location

The workflow is available:

- ☐ As part of the paper's supplementary material.
- ☒ In this Git repository: README.md
- ☐ Other (please specify):

Format(s)

- ☐ Single master code file
- ☐ Wrapper (shell) script(s)
- ☐ Self-contained Quarto, R Markdown file, Jupyter notebook, or other literate programming approach
- ☒ Text file (e.g., a readme-style file) that documents workflow
- ☐ Makefile
- ☐ Other (more detail in *Instructions* below)

Instructions

Procedure to reproduce Figures 1, 2, 3, and S1

Step 1. Run figure/visual_plot.R to reproduce Figures 1, 2, 3, and S1.

Procedure to reproduce simulation results on prediction (recommend using high performance computer)

Step 1. Run simulation/prediction.R to get required statistical model results needed for Figures 4, S3, and S4.

Step 2. Run simulation/python_data_generation.R to generate required simulated data for neural network results needed for Figures 4, S3, and S4.

Step 3. Run simulation/LSTM_sim.ipynb to generate required LSTM model results needed for Figures 4, S3, and S4.

Step 4. Run simulation/TimePFN_sim.ipynb to generate required TimePFN model results needed for Figures 4, S3, and S4.

Step 5. Run simulation/prediction_figure.R to reproduce Figures 4, S3, and S4.

Procedure to reproduce simulation results on estimation (recommend using high performance computer)

Step 1. Run simulation/estimation.R to reproduce Tables S1, S2, and S3.

Procedure to reproduce real data results

Step 1. Run real data/real_data_engerdy.R to reproduce Figures 5, 6, and S2, and to generate required statistical model results needed for Figure S5.

Step 2. Run real data/real_data_trip.R to reproduce Figures 7, 8, S6, S7, S8, S9, and S10, and to generate required statistical model results needed for Figure S11.

Step 3. Run real data/LSTM_real.ipynb to generate required LSTM model results needed for Figures S5 and S11.

Step 4. Run real data/TimePFN_real.ipynb to generate required TimePFN model results needed for Figures S5 and S11.

Step 5. Run real data/prediction_figure.R to reproduce Figures S5 and S11.

Expected run-time

Approximate time needed to reproduce the analyses on a standard desktop machine:

- ☐ < 1 minute
- ☐ 1-10 minutes
- ☐ 10-60 minutes
- ☐ 1-8 hours
- ☐ > 8 hours
- ☒ Not feasible to run on a desktop machine, as described here: the simulation requires high performance computers, while the real data analysis results can be reproduced on a standard desktop machine for around 4 hours.

Additional information (optional)

Notes (optional)